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# Study of degeneracies in pulsar noise analysis

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# Contents

**Introduction** **1**

**1 Gravitational waves** **2**
1.1 Gravitational waves in linearized general relativity . . . . . 2
1.2 Gravitational waves from binary systems . . . . . 3
1.3 Stochastic gravitational wave background . . . . . 5

**2 Pulsar Timing Array** **6**
2.1 Pulsar timing . . . . . 6
2.2 Gravitational waves in pulsar timing . . . . . 8

**3 The pulsar noise model** **11**
3.1 Gaussian processes with power-law spectra . . . . . 11
3.2 White noise . . . . . 12
3.3 Intrinsic red noise . . . . . 12
3.4 Chromatic red noise . . . . . 13
3.4.1 Dispersion measure variations . . . . . 13
3.4.2 Scattering variations . . . . . 14
3.4.3 Other chromatic noise processes . . . . . 14
3.5 Band noise . . . . . 15

**4 Bayesian inference and analysis tools** **16**

**5 Validation of Discovery on PSR J1640+2224** **17**

**6 Study of degeneracies** **18**

**Conclusions** **19**

# Introduction

# Chapter 1

# Gravitational waves

## 1.1 Gravitational waves in linearized general relativity

In general relativity, the geometry of spacetime, described by the metric $g_{\mu\nu}$, is related to the distribution of matter and energy through Einstein's field equations:

$$G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2} R\, g_{\mu\nu} = 8\pi\, T_{\mu\nu}\,, \tag{1.1}$$

where $G_{\mu\nu}$ is the Einstein tensor, $R_{\mu\nu} \equiv R^{\lambda}{}_{\mu\lambda\nu}$ is the Ricci tensor, $R = g^{\mu\nu} R_{\mu\nu}$ is the Ricci scalar, and $T_{\mu\nu}$ is the stress-energy tensor [1].

In the weak-field limit, the metric can be written as a small perturbation of the flat Minkowski spacetime:

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}\,, \qquad |h_{\mu\nu}| \ll 1\,, \tag{1.2}$$

where $\eta_{\mu\nu} = \mathrm{diag}(-1, 1, 1, 1)$ is the Minkowski metric and $h_{\mu\nu}$ is a small tensor perturbation of the metric. To first order in $h_{\mu\nu}$, indices are raised and lowered with $\eta_{\mu\nu}$.

We introduce the trace-reversed perturbation

$$\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu} h\,, \qquad h = \eta^{\mu\nu} h_{\mu\nu}\,. \tag{1.3}$$

Expanding the Einstein tensor to first order in $h_{\mu\nu}$ and using the Lorenz gauge $\partial^{\mu} \bar{h}_{\mu\nu} = 0$, the field equations become

$$\Box\, \bar{h}_{\mu\nu} = -16\pi\, T_{\mu\nu}\,, \tag{1.4}$$

where $\Box = \eta^{\mu\nu} \partial_{\mu} \partial_{\nu}$ is the flat-space d'Alembert operator.

Far from the source, Eq. (1.4) becomes

$$\Box\, \bar{h}_{\mu\nu} = 0\,, \tag{1.5}$$

which is a wave equation whose solutions propagate at the speed of light.

In vacuum, the residual gauge freedom can also be used to impose the transverse-traceless (TT) gauge, defined by

$$h_{0\mu} = 0\,, \qquad h^{i}{}_{i} = 0\,, \qquad \partial^{j} h_{ij} = 0\,. \tag{1.6}$$

In the TT gauge the perturbation is traceless, so that $\bar{h}_{\mu\nu} = h_{\mu\nu}$. For a monochromatic plane wave travelling along the $z$-axis, the solution then takes the form

$$h^{\mathrm{TT}}_{\mu\nu}(t,z) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & h_+ & h_\times & 0 \\ 0 & h_\times & -h_+ & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \cos\left[\omega(t-z)\right]. \tag{1.7}$$

Here $h_+$ and $h_\times$ are the amplitudes of the two independent polarizations of the wave: as for electromagnetic waves, only two polarization states are allowed. Since all the $t$ and $z$ components are zero, the wave only acts on the plane perpendicular to its direction of propagation. The + polarization stretches distances along the $x$-axis while compressing them along the $y$-axis (and vice versa half a period later), whereas the $\times$ polarization produces the same deformation rotated by 45°.

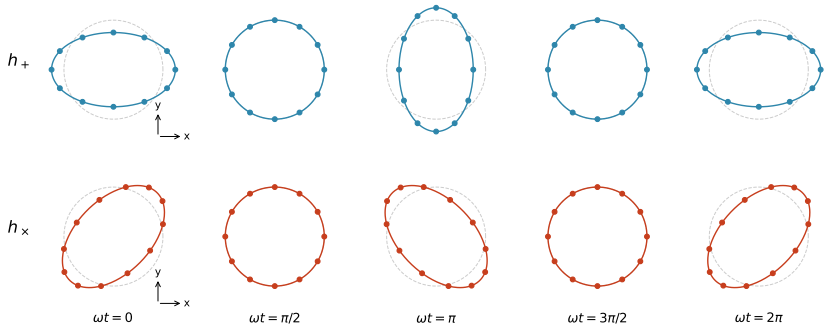



FIGURE 1.1: Deformation of a ring of test masses in the $xy$ plane over one period, for a gravitational wave propagating along $z$ with polarization + (top) and $\times$ (bottom).

## 1.2 Gravitational waves from binary systems

As in electromagnetism, the radiation field of a source can be expanded in multipoles. We consider a source far from the observer and moving slowly ($v \ll c$). The monopole and dipole terms make no contribution to the radiation, as a consequence of stress-energy conservation, which implies the conservation of the total mass-energy and of the total momentum of the system, respectively.

The leading contribution to the radiation is therefore the quadrupole term [1],

$$h_{ij}^{\mathrm{TT}} \sim \frac{2G}{c^4}\frac{\ddot{Q}_{ij}}{r}\,, \qquad Q_{ij} = \int \mathrm{d}^3x\,\rho\left(x_i x_j - \frac{1}{3}\,|\mathbf{x}|^2\delta_{ij}\right), \tag{1.8}$$

where $Q_{ij}$ is the reduced (traceless) quadrupole moment of the mass density $\rho$ of the source.

The most important sources of gravitational waves are binary systems of compact objects, whose orbital motion produces a time-varying quadrupole moment. For a binary on a circular orbit with orbital angular frequency $\omega_s$, the mass distribution returns to the same configuration after half an orbit, so that the quadrupole moment oscillates twice per orbital period. As a consequence, gravitational waves are emitted at twice the orbital frequency,

$$\omega_{\mathrm{gw}} = 2\,\omega_s\,, \qquad \Rightarrow \qquad f_{\mathrm{gw}} = \frac{\omega_{\mathrm{gw}}}{2\pi} = 2f_{\mathrm{orb}}\,. \tag{1.9}$$

Using Kepler's law to eliminate the orbital radius, the amplitude of the emitted wave, averaged over the orientations of the binary, can be written as [2]

$$h = \sqrt{\frac{32}{5}}\,\frac{(G\mathcal{M}_c)^{5/3}\,(\pi f_{\mathrm{gw}})^{2/3}}{c^4\,r}\,, \tag{1.10}$$

where $r$ is the distance to the source and $\mathcal{M}_c = (m_1 m_2)^{3/5}/(m_1+m_2)^{1/5}$ is the chirp mass of the binary, with $m_1$ and $m_2$ the masses of the two components. At this order, the amplitude depends on the masses only through the combination $\mathcal{M}_c$.

The total energy of the orbit can be written as a function of the gravitational-wave frequency,

$$E_{\mathrm{orb}} = -\frac{1}{2}\,\mathcal{M}_c\,(\pi G\mathcal{M}_c f_{\mathrm{gw}})^{2/3}\,, \tag{1.11}$$

while the power radiated in gravitational waves, obtained from the quadrupole formula, is

$$P = \frac{32}{5}\,\frac{c^5}{G}\left(\frac{\pi G\mathcal{M}_c f_{\mathrm{gw}}}{c^3}\right)^{10/3}\,. \tag{1.12}$$

Since the energy carried away by the waves is known from the orbit, $P = -\mathrm{d}E_{\mathrm{orb}}/\mathrm{d}t$, combining these two equations gives the evolution of the frequency,

$$\dot{f}_{\mathrm{gw}} = \frac{96}{5}\,\pi^{8/3}\left(\frac{G\mathcal{M}_c}{c^3}\right)^{5/3} f_{\mathrm{gw}}^{11/3}\,. \tag{1.13}$$

We note that binaries emitting at low frequencies evolve very slowly.

Since the characteristic frequency of the emission scales inversely with the total mass of the system, the binaries emitting at the lowest frequencies are supermassive black hole binaries (SMBHBs), with masses of $10^8$–$10^{10}\,M_\odot$. Supermassive black holes reside at the centres of most galaxies and cannot be

the product of stellar evolution. It is widely accepted that SMBHBs form as a natural consequence of galaxy mergers: when two galaxies merge, their central black holes form a binary system which, once it reaches separations of the order of milliparsecs, emits gravitational waves in the nanohertz band [2, 3].

## 1.3 Stochastic gravitational wave background

A large population of compact binaries, such as SMBHBs, emitting at comparable frequencies and with comparable amplitudes cannot be resolved individually. The resulting signal is an incoherent superposition of the individual waves, which forms a stochastic gravitational wave background (GWB). A GWB is defined by the fact that the signal $h(t)$ is a random process, which cannot be described by a deterministic waveform but only statistically, through its ensemble averages. We will make the following assumptions on the GWB [1]:

- **stationary**: its statistical properties do not change in time;
- **Gaussian**: it is fully characterized by its two-point correlator;
- **isotropic**: its statistical properties are the same in every direction;
- **unpolarized**: the two polarizations are uncorrelated and have the same intensity.

Under these assumptions, the background is completely characterized by a single function of frequency, its one-sided power spectral density (PSD) $S_h(f)$, which describes how the power of the signal is distributed in frequency. From $S_h(f)$ it is convenient to define the characteristic strain [2]

$$h_c(f) = \sqrt{f\, S_h(f)}\,, \tag{1.14}$$

a dimensionless quantity that can be directly compared with the strain amplitudes of individual sources discussed in Section 1.2.

Assuming that all binaries are on circular orbits and that their evolution is driven only by the emission of gravitational waves, the energy radiated per logarithmic frequency interval follows from Eqs. (1.12) and (1.13), $f_{\rm gw}\, P/\dot{f}_{\rm gw} \propto f_{\rm gw}^{2/3}$. Summing over the whole population, the characteristic strain spectrum of the background follows the power law [2, 4]

$$h_c(f) = A_{\rm GWB} \left(\frac{f}{f_{\rm yr}}\right)^{-2/3}, \tag{1.15}$$

where $A_{\rm GWB}$ is the amplitude at the reference frequency $f_{\rm yr} = 1\,{\rm yr}^{-1}$. While the spectral index is fixed by the physics of the individual binaries, the amplitude depends on the properties of the SMBHB population, and is of the order of $10^{-15}$ [5–8]. In Chapter 2 we will see how this strain spectrum translates into a red noise process in the timing residuals, common to all pulsars.

# Chapter 2

# Pulsar Timing Array

Pulsar timing arrays (PTAs) [9–11] are currently the most sensitive method for detecting gravitational waves in the nanohertz band, where the signal is expected to be dominated by the population of supermassive black hole binaries (SMBHBs) discussed in Section 1.2.

## 2.1 Pulsar timing

The PTA technique relies on pulsars, which are rapidly rotating and highly magnetized neutron stars [12] that emit beams of radio radiation along their magnetic axis. Pulsars are commonly classified according to their rotational period into classical pulsars, with periods of the order of one second, and millisecond pulsars (MSPs) [13], with periods of a few milliseconds. MSPs have a much more stable rotation, which allows the times of arrival (TOAs) of their pulses to be measured with higher precision.

The measurement of the TOAs involves several steps [2, 14]. The radio emission of the pulsar propagates through the ionized interstellar medium (ISM), whose refractive index depends on frequency. As a consequence, the signal suffers a frequency-dependent delay proportional to $\nu^{-2}$, where $\nu$ is the observing radio frequency, so that higher radio frequencies arrive earlier than lower ones. This dispersion can be corrected in two ways. In incoherent dedispersion, the observed band is divided into narrow frequency channels, and each channel is shifted in time according to the $\nu^{-2}$ law before summing them, so that the pulse is aligned across the whole band. In coherent dedispersion, the raw signal is instead convolved with the inverse transfer function of the ISM, which removes the dispersive delay exactly.

Individual pulses are usually too weak to be distinguished from the background noise. To overcome this problem, the data are divided into segments of length equal to the pulsar period and summed together, a procedure known as folding. Since the pulsed signal always falls at the same rotational phase, it adds up coherently, while the noise averages out, so that the signal-to-noise ratio grows as the square root of the number of folded pulses. The result is an average pulse profile which, unlike individual pulses, is remarkably stable over time, and which is used to determine the TOA.

The TOA is then obtained by cross-correlating the folded profile with a high signal-to-noise template of the same pulsar. The resulting phase shift, converted into time and added to the reference time of the observation, gives the arrival time of the pulse together with its uncertainty.

Since the TOAs are measured at the observatory, which is not an inertial reference frame, they are then referred to the Solar System barycenter (SSB) through a series of clock and propagation corrections. The barycentric TOAs are finally compared with the predictions of a timing model, which describes the rotation of the pulsar together with its astrometric and, for binary systems, orbital parameters. In this work we use the timing solution of the EPTA DR2low dataset [15], which combines the EPTA second data release [16] with low-frequency observations from LOFAR. It was obtained by refitting the EPTA timing model to the combined data with software based on TEMPO2 [17, 18]. The difference between the observed and the predicted TOAs defines the timing residuals,

$$\delta t_i = t^{\mathrm{obs}}_{\mathrm{TOA},i} - t^{\mathrm{model}}_{\mathrm{TOA},i}. \tag{2.1}$$

Any effect not included in the timing model, such as noise processes or gravitational waves, leaves a signature in the residuals. Since the TOAs are typically collected every few weeks over a time span $T$ of one or two decades, PTAs are sensitive to signals with frequencies between about $1/T$, which is 1.6 nHz for $T = 20\,\mathrm{yr}$, and the Nyquist frequency $1/(2\Delta t)$ set by the observing cadence $\Delta t$, which is about $4 \times 10^{-7}\,\mathrm{Hz}$ for observations every two weeks [19]. As an example, Figure 2.1 shows the timing residuals of PSR J1640+2224, the pulsar analyzed in this work, in the DR2low dataset. The residuals at low radio frequencies have a much larger scatter, partly because of the larger TOA uncertainties and partly because the effects of the ISM are stronger at low frequencies, as discussed in Chapter 3.

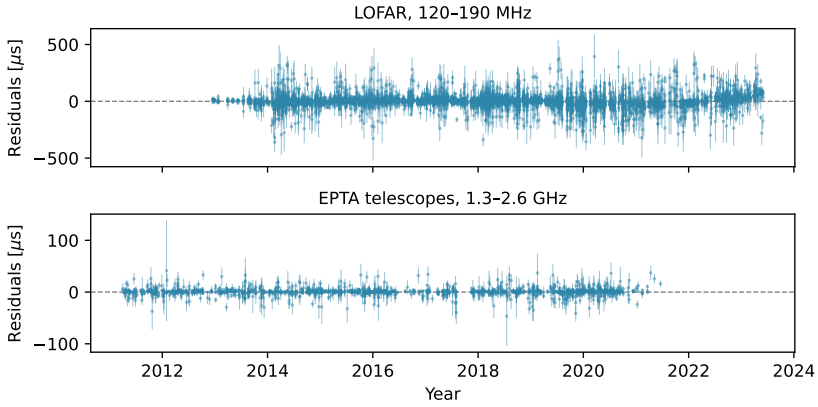



FIGURE 2.1: Timing residuals of PSR J1640+2224 in the EPTA DR2low dataset [15], for the LOFAR observations at 120–190 MHz (top) and for the EPTA telescopes at 1.3–2.6 GHz (bottom). Note the different vertical scales.

## 2.2 Gravitational waves in pulsar timing

A gravitational wave passing between the Earth and a pulsar perturbs the spacetime metric along the path of the radio pulses, slightly changing their travel time. As a result, the rate at which the pulses are received is shifted with respect to the intrinsic rotational frequency of the pulsar. This effect is described by the redshift [2, 10, 20]

$$z(t) \equiv \frac{\nu_0 - \nu(t)}{\nu_0}, \tag{2.2}$$

where $\nu_0$ is the unperturbed pulse frequency and $\nu(t)$ the observed one. Since the TOAs accumulate this shift over time, the contribution of the gravitational wave to the timing residuals is given by its time integral,

$$R(t) = \int_0^t z(t')\, dt'. \tag{2.3}$$

For a stochastic background, $R(t)$ is itself a random process and cannot be predicted as a function of time. It is instead characterized by its power spectral density, which follows from the strain spectral density $S_h(f)$ introduced in Section 1.3. Two factors relate the two quantities. Averaging the response of the Earth–pulsar system over all propagation directions of the background gives a factor $1/3$, while the time integral in Eq. (2.3) introduces a factor $1/(2\pi f)^2$. Using Eq. (1.14), the PSD of the timing residuals induced by the GWB is [2]

$$S(f) = \frac{h_c^2(f)}{12\pi^2 f^3}. \tag{2.4}$$

Inserting the power-law spectrum of Eq. (1.15), expected for a population of circular SMBHBs whose evolution is driven by GW emission, we obtain

$$S(f) = \frac{A_{\rm GWB}^2}{12\pi^2} \left(\frac{f}{f_{\rm yr}}\right)^{-\gamma_{\rm GWB}} f_{\rm yr}^{-3}, \qquad \gamma_{\rm GWB} = \frac{13}{3}, \tag{2.5}$$

since $h_c^2/f^3 \propto f^{-4/3}\, f^{-3} = f^{-13/3}$. This PSD is a power law with the same functional form used to describe the intrinsic red noise of a pulsar. As a consequence, in the timing residuals of a single pulsar the GWB is indistinguishable from intrinsic red noise. The signal can only be identified by exploiting the fact that it is common to all the pulsars of the array and induces a specific correlation between their residuals.

The correlation between the timing residuals of two pulsars $i$ and $j$ is described by the Hellings–Downs curve [21]. Under the assumptions introduced in Section 1.3, this correlation does not depend on the positions of the pulsars on the sky, but only on their angular separation $\zeta_{ij}$,

$$\Gamma_{ij} = \frac{3}{2}\, x_{ij} \ln x_{ij} - \frac{x_{ij}}{4} + \frac{1}{2} + \frac{1}{2}\,\delta_{ij}, \qquad x_{ij} = \frac{1 - \cos\zeta_{ij}}{2}, \tag{2.6}$$

where $\delta_{ij}$ is the Kronecker delta. The cross-PSD of the residuals of pulsars $i$ and $j$ is then $S_{ij}(f) = \Gamma_{ij}\, S(f)$, with $S(f)$ given by Eq. (2.4), so that for a single pulsar ($\Gamma_{ii} = 1$) Eq. (2.4) is recovered.

The Hellings–Downs curve is shown in Figure 2.2. The correlation tends to 1/2 for pulsars that are close on the sky, ~~reaches a minimum of about $-0.15$ at $\zeta \simeq 82°$,~~ and rises again to 1/4 for pulsars in opposite directions. This angular pattern reflects the quadrupolar nature of gravitational waves and is the distinctive signature of a GWB.

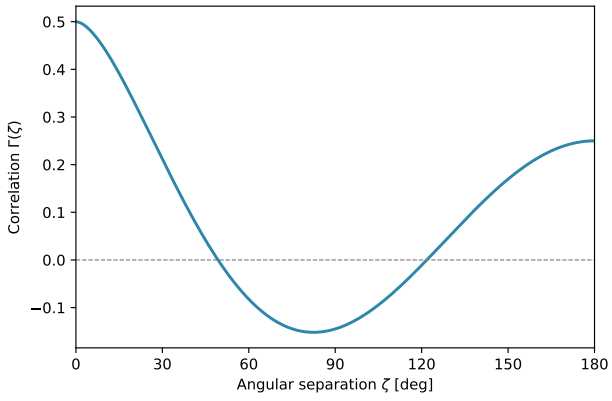



FIGURE 2.2: The Hellings–Downs curve of Eq. (2.6), giving the expected correlation between the timing residuals of two distinct pulsars as a function of their angular separation $\zeta$.

In 2023 four PTA collaborations, EPTA with InPTA, NANOGrav, PPTA

and CPTA, reported evidence for a common red process with Hellings–Downs correlations [5–8]. Since the signal is weak and can be mimicked by the red noise of each pulsar, an accurate description of the pulsar noise is essential. The noise sources and their modeling are described in Chapter 3.

# Chapter 3

# The pulsar noise model

The TOAs can be described as the sum of a deterministic contribution, given by the timing model introduced in Section 2.1, and a stochastic contribution due to several noise processes [19, 22]. In this chapter we describe these noise processes and the way they are modeled. Although they have different physical origins, they are all modeled as Gaussian processes.

A first broad distinction is between white noise and red noise. White noise has a flat power spectral density, that is, the same power at all Fourier frequencies $f$ of the timing residuals, so it produces no slow trends in the residuals. Red noise instead has more power at low frequencies, and it produces slow, time-correlated variations of the residuals [2]. When two noise processes have a similar effect on the timing residuals, the data cannot fully separate them, and part of the power of one process can be attributed to the other. Their parameters are then degenerate [19, 23].

## 3.1 Gaussian processes with power-law spectra

All the red noise processes considered in this work are described in the same way. A red noise process in the timing residuals is written as a Fourier series truncated to $N$ frequencies [2],

$$\delta t_{\mathrm{red}}(t_i) = \sum_{k=1}^{N} \Big[ a_k \sin(2\pi f_k t_i) + b_k \cos(2\pi f_k t_i) \Big], \qquad f_k = \frac{k}{T}, \tag{3.1}$$

where $T$ is the time span of the data. The lowest frequency is therefore $1/T$, while the number of components $N$ sets the highest one, $N/T$. Only low frequencies are needed, since red noise has most of its power there.

The coefficients $a_k$ and $b_k$ are independent Gaussian random variables with zero mean, whose variance is fixed by the PSD of the process,

$$\langle a_k^2 \rangle = \langle b_k^2 \rangle = \frac{S(f_k)}{T}. \tag{3.2}$$

For the PSD we use a power law with amplitude $A$ and spectral index $\gamma$,

$$S(f) = \frac{A^2}{12\pi^2} \left(\frac{f}{f_{\rm yr}}\right)^{-\gamma} f_{\rm yr}^{-3}, \tag{3.3}$$

which for $\gamma = \gamma_{\rm GWB} = 13/3$ gives back the PSD of the GWB in Eq. (2.5). Each red noise process is therefore described by only two parameters, $A$ and $\gamma$.

## 3.2 White noise

As mentioned above, white noise has a flat power spectral density. In pulsar timing analysis there are three main sources of white noise [2], each described by a parameter defined separately for every observing system, that is, every combination of receiver and backend.

- **Radiometer noise (EFAC).** The TOA uncertainty $\sigma_{{\rm TOA},i}$ obtained from the template fit may not include all sources of error, so it is multiplied by a correction factor, the EFAC.
- **Additional white noise (EQUAD).** White noise not related to the radiometer, for example due to instrumental effects, is added in quadrature to the TOA uncertainty through the EQUAD.
- **Pulse jitter (ECORR).** Since each TOA is obtained by folding a finite number of pulses, the folded profile fluctuates from one observation to the next. This noise is fully correlated between TOAs of the same observing epoch and uncorrelated between different epochs, and it is described by the ECORR.

Without ECORR, the variance of the $i$-th TOA is then

$$\sigma_i^2 = {\rm EFAC}^2\, \sigma_{{\rm TOA},i}^2 + {\rm EQUAD}^2. \tag{3.4}$$

In this work the white noise parameters are fixed to the values provided with the DR2low dataset, and ECORR is included only for the LOFAR systems [15].

## 3.3 Intrinsic red noise

The intrinsic red noise, also called spin noise, results from irregularities in the rotation of the pulsar [19]. Although MSPs have a very stable rotation, they still show small fluctuations in their rotational phase and period [2]. These fluctuations produce a red noise process that is achromatic, that is, it does not depend on the observing radio frequency $\nu$. Its power spectral density is

modeled as a power law,

$$S_{\mathrm{RN}}(f) = \frac{A_{\mathrm{RN}}^2}{12\pi^2}\left(\frac{f}{f_{\mathrm{yr}}}\right)^{-\gamma_{\mathrm{RN}}} f_{\mathrm{yr}}^{-3}, \tag{3.5}$$

where $A_{\mathrm{RN}}$ is the amplitude at the reference frequency $f_{\mathrm{yr}}$ and $\gamma_{\mathrm{RN}}$ is the spectral index. This PSD has the same functional form as the one of the GWB in Eq. (2.5). As discussed in Section 2.2, in a single pulsar the intrinsic red noise therefore cannot be distinguished from the GWB.

~~The inferred parameters of the intrinsic red noise also depend on how the other noise processes are modeled. For example, NANOGrav and PPTA, which describe the DM variations with different methods, find slightly different spin noise parameters for the pulsars observed by both arrays [19].~~

## 3.4 Chromatic red noise

The radio pulses reach the Earth after travelling through the ionized interstellar medium, which delays them by an amount that depends on the observing radio frequency $\nu$, with lower frequencies delayed more. Since the pulsar, the Earth and the ISM move with respect to each other, the electron content along the line of sight changes with time, and so do these delays [19]. The result is a red noise process in the timing residuals whose amplitude depends on $\nu$, called chromatic red noise.

Chromatic red noise is modeled with the same Fourier basis used for the achromatic red noise, scaled by a power of the radio frequency [15, 23],

$$\delta t_{\mathrm{chrom}}(t_i, \nu_i) = \left(\frac{\nu_{\mathrm{ref}}}{\nu_i}\right)^{\alpha} \sum_{k=1}^{N} \Big[a_k \sin(2\pi f_k t_i) + b_k \cos(2\pi f_k t_i)\Big], \tag{3.6}$$

where $\nu_i$ is the radio frequency of the $i$-th TOA, $\nu_{\mathrm{ref}} = 1400\,\mathrm{MHz}$ is a reference frequency and $\alpha$ is the chromatic index. The case $\alpha = 0$ gives back the achromatic red noise, while $\alpha = 2$ and $\alpha = 4$ describe the dispersion measure variations and the scattering variations discussed below. Because of the factor $\nu_i^{-\alpha}$, these processes are much stronger at low radio frequencies.

### 3.4.1 Dispersion measure variations

As mentioned in Section 2.1, a radio wave travelling through the ionized ISM is delayed by an amount that depends on its frequency. For a cold plasma the dispersive delay is

$$\Delta t_{\mathrm{DM}} = \mathcal{D}\,\frac{\mathrm{DM}}{\nu^2}, \qquad \mathrm{DM} = \int_0^L n_e \,\mathrm{d}l, \tag{3.7}$$

where the dispersion measure DM is the integral of the electron density $n_e$ along the line of sight, usually expressed in $\mathrm{pc\,cm^{-3}}$, and $\mathcal{D} = 4.148808 \times 10^3\,\mathrm{s\,MHz^2\,cm^3\,pc^{-1}}$ is the dispersion constant [19].

The mean value of the DM and its linear and quadratic trends in time are included in the timing model [2]. Because of the relative motion of the pulsar, the Earth and the ISM, however, the line of sight crosses different electron densities over time, and the remaining DM variations are stochastic. They are modeled as a chromatic red noise process, Eq. (3.6) with $\alpha = 2$, with a power-law PSD of the same form as the intrinsic red noise,

$$S_{\mathrm{DM}}(f) = \frac{A_{\mathrm{DM}}^2}{12\pi^2} \left(\frac{f}{f_{\mathrm{yr}}}\right)^{-\gamma_{\mathrm{DM}}} f_{\mathrm{yr}}^{-3}, \tag{3.8}$$

where $A_{\mathrm{DM}}$ is the amplitude of the delay at the reference frequency $\nu_{\mathrm{ref}}$.

### 3.4.2 Scattering variations

Inhomogeneities in the ISM also scatter the radio waves, so that part of the signal reaches the observer along longer paths. This multipath propagation broadens the pulse with an exponential tail, whose characteristic timescale $\tau_{\mathrm{sc}}$ depends strongly on the radio frequency [19],

$$\tau_{\mathrm{sc}} \propto \nu^{-\alpha}. \tag{3.9}$$

Since the line of sight changes with time, the scattering also varies and introduces an additional chromatic red noise in the TOAs. It is modeled with Eq. (3.6) and a power-law PSD of the same form as Eq. (3.8), with amplitude $A_{\mathrm{SV}}$ and spectral index $\gamma_{\mathrm{SV}}$.

The value of $\alpha$ depends on the properties of the scattering medium. A thin screen with inhomogeneities of the same size gives $\alpha = 4$, which is the value adopted for the DR2low dataset, while an extended medium with Kolmogorov turbulence gives $\alpha = 4.4$ [15]. In this work we use $\alpha = 4$ as the reference model, and we study the effect of $\alpha = 4.4$ and of a free chromatic index in Chapter 6.

### 3.4.3 Other chromatic noise processes

DM variations and scattering are not the only chromatic effects in pulsar timing. The solar wind adds a further contribution to the DM, which is largest when the line of sight passes close to the Sun and therefore varies along the year. Other effects include sudden chromatic events, such as exponential dips in the residuals, and additional delays caused by refraction in the ISM [15, 19]. For PSR J1640+2224, which lies far from the ecliptic plane, the solar wind is described only by a deterministic term in the timing model [15], and none

~~of these processes is included in the noise model. The only additional noise process considered in this work is the band noise, discussed in Section 3.5.~~

## 3.5 Band noise

Band noise is a red noise process that is present only in the TOAs observed within a given radio frequency band [22]. In this work the band considered is the LOFAR band, 120–190 MHz, which contains the TOAs recorded by the seven LOFAR observing systems of the DR2low dataset.

The physical origin of band noise is not known with certainty. It may come from effects of the ISM that are not coherent between different bands, for example because photons at different frequencies cross different scattering volumes, or that scale with frequency more steeply than $\nu^{-2}$, and therefore appear as an excess of noise at low frequencies. Alternatively, it may be caused by frequency-dependent calibration errors, which can remain correlated over hundreds of days, or by radio frequency interference (RFI) present in the same band at different sites. RFI, however, is expected to be mostly uncorrelated between observations, and in that case it is absorbed by the white noise parameters. In the first IPTA data release, band noise was detected in four pulsars, with amplitudes $\log_{10} A_{\mathrm{BN}}$ between about $-14$ and $-12$ and spectral indices $\gamma_{\mathrm{BN}}$ between about 0.6 and 3.3 [22].

Band noise must be distinguished from system noise, a red noise process that affects only the TOAs of a single observing system, that is, a single combination of telescope, receiver and backend, and that can arise from instrumental effects or calibration errors. If a band contains only one system, the two models act on exactly the same TOAs and are completely covariant, so that the data cannot distinguish between them. When several systems observe in the same band, as for the seven LOFAR systems used here, a noise process common to all of them can instead be attributed to the band. ~~For this reason, low frequency observations such as those of LOFAR, together with wide band receivers, are particularly useful to identify band dependent effects [22].~~

The band noise is modeled as a power law of the same form as Eq. (3.5), with amplitude $A_{\mathrm{BN}}$ and spectral index $\gamma_{\mathrm{BN}}$. The Fourier basis of Section 3.1 is applied only to the TOAs of the LOFAR band and is set to zero for all the other TOAs. Unlike the chromatic noise of Eq. (3.6), no factor $(\nu_{\mathrm{ref}}/\nu_i)^{\alpha}$ is included, since the dependence on the radio frequency enters only through the selection of the TOAs. We use $N = 50$ Fourier components. This number was fixed a priori and was not obtained from a model selection.

# Chapter 4

# Bayesian inference and analysis tools

# Chapter 5

# Validation of Discovery on PSR J1640+2224

# Chapter 6

# Study of degeneracies

# Conclusions

# Bibliography

[1] Michele Maggiore. *Gravitational Waves. Volume 1: Theory and Experiments*. Oxford University Press, Oxford, 2007. doi: 10.1093/acprof:oso/9780198570745.001.0001.

[2] Stephen R. Taylor. The nanohertz gravitational wave astronomer, 2021.

[3] A. Sesana, A. Vecchio, and C. N. Colacino. The stochastic gravitational-wave background from massive black hole binary systems: implications for observations with Pulsar Timing Arrays. *Monthly Notices of the Royal Astronomical Society*, 390(1):192–209, 2008. doi: 10.1111/j.1365-2966.2008.13682.x.

[4] E. S. Phinney. A practical theorem on gravitational wave backgrounds, 2001.

[5] EPTA Collaboration and InPTA Collaboration, J. Antoniadis, et al. The second data release from the European Pulsar Timing Array. III. Search for gravitational wave signals. *Astronomy & Astrophysics*, 678:A50, 2023. doi: 10.1051/0004-6361/202346844.

[6] G. Agazie et al. The NANOGrav 15 yr data set: Evidence for a gravitational-wave background. *The Astrophysical Journal Letters*, 951 (1):L8, 2023. doi: 10.3847/2041-8213/acdac6.

[7] D. J. Reardon et al. Search for an isotropic gravitational-wave background with the Parkes Pulsar Timing Array. *The Astrophysical Journal Letters*, 951(1):L6, 2023. doi: 10.3847/2041-8213/acdd02.

[8] H. Xu et al. Searching for the nano-hertz stochastic gravitational wave background with the Chinese Pulsar Timing Array data release I. *Research in Astronomy and Astrophysics*, 23(7):075024, 2023. doi: 10.1088/1674-4527/acdfa5.

[9] M. V. Sazhin. Opportunities for detecting ultralong gravitational waves. *Soviet Astronomy*, 22:36–38, 1978.

[10] S. Detweiler. Pulsar timing measurements and the search for gravitational waves. *The Astrophysical Journal*, 234:1100–1104, 1979. doi: 10.1086/157593.

[11] R. S. Foster and D. C. Backer. Constructing a pulsar timing array. *The Astrophysical Journal*, 361:300–308, 1990. doi: 10.1086/169195.

[12] A. Hewish, S. J. Bell, J. D. H. Pilkington, P. F. Scott, and R. A. Collins. Observation of a rapidly pulsating radio source. *Nature*, 217:709–713, 1968. doi: 10.1038/217709a0.

[13] D. C. Backer, S. R. Kulkarni, C. Heiles, M. M. Davis, and W. M. Goss. A millisecond pulsar. *Nature*, 300:615–618, 1982. doi: 10.1038/300615a0.

[14] D. R. Lorimer and M. Kramer. *Handbook of Pulsar Astronomy*, volume 4 of *Cambridge Observing Handbooks for Research Astronomers*. Cambridge University Press, Cambridge, 2005.

[15] F. Iraci, A. Chalumeau, C. Tiburzi, J. P. W. Verbiest, A. Possenti, et al. Combining the second data release of the European Pulsar Timing Array with low-frequency pulsar data. *Astronomy & Astrophysics*, 704:A109, 2025. doi: 10.1051/0004-6361/202555516.

[16] EPTA Collaboration, J. Antoniadis, et al. The second data release from the European Pulsar Timing Array. I. The dataset and timing analysis. *Astronomy & Astrophysics*, 678:A48, 2023. doi: 10.1051/0004-6361/202346841.

[17] G. B. Hobbs, R. T. Edwards, and R. N. Manchester. TEMPO2, a new pulsar-timing package – I. An overview. *Monthly Notices of the Royal Astronomical Society*, 369(2):655–672, 2006. doi: 10.1111/j.1365-2966.2006.10302.x.

[18] R. T. Edwards, G. B. Hobbs, and R. N. Manchester. TEMPO2, a new pulsar timing package – II. The timing model and precision estimates. *Monthly Notices of the Royal Astronomical Society*, 372(4):1549–1574, 2006. doi: 10.1111/j.1365-2966.2006.10870.x.

[19] G. M. Shaifullah. Pulsar timing arrays—challenges, and current status. *Classical and Quantum Gravity*, 42(24):243001, 2025. doi: 10.1088/1361-6382/ae19c2.

[20] Michele Maggiore. *Gravitational Waves. Volume 2: Astrophysics and Cosmology*. Oxford University Press, Oxford, 2018. doi: 10.1093/oso/9780198570899.001.0001.

[21] R. W. Hellings and G. S. Downs. Upper limits on the isotropic gravitational radiation background from pulsar timing analysis. *The Astrophysical Journal Letters*, 265:L39–L42, 1983. doi: 10.1086/183954.

[22] L. Lentati, R. M. Shannon, W. A. Coles, et al. From spin noise to systematics: stochastic processes in the first International Pulsar Timing Array

data release. *Monthly Notices of the Royal Astronomical Society*, 458(2): 2161–2187, 2016. doi: 10.1093/mnras/stw395.

[23] I. Ferranti, M. Falxa, A. Sesana, A. Chalumeau, N. Porayko, G. Shaifullah, I. Cognard, L. Guillemot, M. Kramer, K. Liu, and G. Theureau. Impact of the observation frequency coverage on the significance of a gravitational wave background detection in pulsar timing array data. *Astronomy & Astrophysics*, 694:A38, 2025. doi: 10.1051/0004-6361/202452805.